Journal of Behavioral and Experimental Economics 54 (2015) 105–117

Contents lists available at ScienceDirect

Journal of Behavioral and Experimental Economics

journal homepage: www.elsevier.com/locate/socec

A prospect theory approach to assessing changes in parameters of insurance contracts with an application to money-back guarantees

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ARTICLE INFO

Article history:
Received 17 October 2013
Revised 21 August 2014
Accepted 22 November 2014
Available online 10 December 2014

Keywords:
Risk
Prospect-theory
Information
Money-back guarantees
Disappointment
Loss-aversion

ABSTRACT

Prospect theory has changed the way economists think about decision making under uncertainty – yet after so many years there have been few applications of the theory and those appearing mostly in finance. One of the barriers to applying the prospect theory is that it is not designed to be applicable (Barberis, 2013). This study applies prospect theory to the selection of money back guarantee (MBG) contracts. When consumers can choose from a menu of MBG contracts they are basically trading off risk with price in a way that resembles a choice of lotteries with multidimensional outcomes. Our application, which integrates reference based utility models with elements of prospect theory and the disappointment model, helps in explaining the large premium attached to MBG contracts that cannot be explained by the expected utility framework. We further show that the combination of probability weighting with disappointment aversion appears to provide a better explanation for consumers' high valuation of MBGs relative to each one when measured separately. We empirically test how consumers' valuation of the MBG option is affected by MBG duration, variation in the likelihood of returns, and return conditions that affect consumers' return cost. Our approach can be applied to model choices of risk reduction mechanisms such as extended warranties, demonstrations, and sampling.
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1. Introduction

Empirical assessments of warranties suggest that they are overvalued by consumers beyond the degree warranted by the expected utility approach. For example, the annual costs of extended warranty contracts for household appliances exceed the average expected costs of repair by considerably more than what might be attributable to profit and overhead (Chen, Kalra and Sun, 2009). Consumers pay 15–25% premiums for 2-year, and 20–30% premiums for 3-year extended warranties. The large difference between these premiums and the expected value of repairs covered by the extended warranty contracts is inconsistent with expected benefit calculations and has been typically considered to reflect consumer judgment errors and asymmetric information (Bryant and Gerner, 1978; Center for Policy Alternatives, 1978).

In this paper we explore whether a similar overvaluation exists of another risk-reduction mechanism, the money-back guarantee (MBG). MBGs offer consumers the option to return a purchased product for a refund if the consumer is dissatisfied with the purchase for whatever reason. The U.S. National Retail Federation (NRF), estimated that 8.7% of purchases are returned under MBG, reflecting a total value of $174 billion of product returns, while Ofek, Katona and Sarvary (2010) estimate that 4–6% of products is returned. Anderson, Hansen and Simester (2009), who analyzed market data taken from a mail-order catalog company, calculate that MBGs had the effect of increasing sales by 53%, 16%, and 9% respectively for women's shoes, women's fashion items and men's apparel. They explain these variations in demand effects as reflecting differences in risk such that women shoes are more risky than the other two categories. In this paper we estimate consumer's willingness to pay for MBGs in one product category to see if these values represent exaggerated premiums for reducing risks in the face of uncertainty from an expected utility framework. Finding that indeed such exaggerated premiums exist, we evaluate whether a weighted value function, disappointment aversion or both might explain these risk reduction premiums.

There are two distinct approaches for incorporating psychological response to losses and gains into models of decision making. One approach incorporates expected losses and gains into the framework

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http://dx.doi.org/10.1016/j.socec.2014.11.006
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of expected utility theory by decomposing the utility into that of consumption and the utility from loss or gain (Bell, 1982; Gul, 1991). Embedded in this approach is the assumption that people maximize their expected utility and that disappointment (happiness) from an outcome that falls (is higher) from a reference level is a separable argument that is concave in gains and convex in losses (Kőszegi and Rabin, 2006, 2007, 2009; Shalev, 2000). The second approach deviates from the expected utility (EU) framework, suggesting that people do not really maximize expected utility, but are instead guided by their (limited) cognitive system. Perhaps the most prominent model of this approach is prospect theory (Kahneman and Tversky, 1979). The core assumptions of the prospect theory model are that outcomes are compared to a certain reference point, which is usually zero, and that the absolute subjective value of losses (i.e., values that fall short of the reference point) is larger than the subjective value of equivalent gains while extended expected utility models allow for non-zero multiple reference points. The cost that is attributed to losses depends on their context (Ert and Erev, 2013) and they are likely to increase when they are attributed to making mistakes (Chua et al., 2009).

In the current paper we develop a prospect theory framework for assessing consumers' valuation of a popular risk-reduction mechanism: the money back guarantee (MBG), considering product returns and the loss from being stuck with a purchased product that turns out to not fit the consumer's needs (non-fit) as disappointment. Prospect theory was originally designed to describe risky choices between two-outcome gambles, and as such it evaluates the subjective values of monetary outcomes and their probabilities (Kahneman and Tversky, 1979). It was later generalized to capture choice between multiple outcome gambles (Tversky and Kahneman, 1992; Wakker, 1994), but has not been used to explain consumers' choice of MBG, perhaps because it is difficult to apply (Barberis, 2013). In order to fill this void we develop an extension of the prospect theory model that can assess discrete choices and choices between continuous alternatives at the same time and relates to outcomes that are more complex than win–lose scenarios. For example, our model allows for one state where the uncertainty is resolved and the benefit (loss) is realized, and another where the uncertainty has not been resolved yet. The analysis allows us to assess how the premium for various risk management contracts depends on a contract's characteristics as well as on mental costs.

While we apply the methodology to MBGs, its applicability is much wider and can be used to estimate the value of various tools that reduce consumer risk, such as warranties, price guarantees, second hand markets, product sampling, leasing, trade-ins and product demonstrations.

MBGs are ex post guarantees that provide the buyer a put option to return the purchased product for any reason. The standard MBG in the U.S. is a 30-day guarantee that enables customers to return the product for any reason for a full refund of the purchase price. In the EU the law specifies 14 days for online purchases while the refund policy of goods bought in stores varies across stores and countries. In addition, the EU retailer bears the shipping costs of returns, while in the U.S. shipping costs are generally paid by the consumer. There are many variations of MBGs that include partial MBGs (the buyer pays restocking fees and or shipping costs for returned products), time varying MBGs (the restocking fees are increasing with the duration of the return), and lifetime MBGs (the guarantee has no time limit, e.g. Nordstrom) and returns being allowed only for products that have not been used (in the EU).

Studies of MBGs as risk management strategies have modeled consumers' choices between buying with and without an MBG, employing either the expected net benefits (Davis, Gerstner and Hagerty, 1995; Heiman et al., 2002; Matthews and Persico, 2005) or expected utility of benefits (Che, 1996) models. In particular, consumers' benefit from an MBG is assumed to be the reimbursement of the product price if it is returned, while the costs of the guarantee are the consumers' return costs if the product is returned, and the price premium paid for bundling the MBG option with the product if the product is not returned. It is typically assumed that the duration of the MBG contract is long enough to resolve fit uncertainty—an assumption that has not been validated yet empirically. Notably, this approach does not account for the consumer's psychological costs of anticipating loss such as the disappointment and regret when they consider the forgone alternative (Loomes and Sugden, 1982) that can result from either buying too short of an MBG if the product did not deliver the expected benefit, or from paying too much if a long MBG contract was purchased and the product provides high benefit. Our proposed extension of the prospect theory framework enables us to estimate and explain consumers' valuation of the MBG.

In this paper we develop an extension of a prospect theory framework that can assess multiple dimensions that vary in nature: some discrete (e.g., whether or not the product fits consumer's idiosyncratic needs) and others continuous (e.g., duration of the return period). The analysis further allows us to assess how the premium for various MBG contracts depends on the contract characteristics and other parameters (return costs). We use this approach to model the MBG contract and develop testable hypotheses about the parameters of interest. From survey data, we calculate individuals' willingness-to-pay premiums (or accept discounts) for better (or worse) MBG terms and estimate the implicit loss from disappointment. The value of disappointment is interpreted here as an extra loss, in addition to the product price, that a consumer experiences when she is stuck with a product that does not fit her needs and this poor fit is discovered only after the MBG expired. Thus the reference point for a consumer buying a given product is the expectation that the product will fit her needs, and the unexpected ex-post realization that these needs were not satisfied generates a psychological sense of loss which can only be eliminated if the consumer receives a reimbursement when the product is returned under an MBG. Our modeling of disappointment as a separable component in the consumer's benefit is consistent with the aforementioned literature. Disappointment is integrated in our model together with the notion of a weighting function drawn from Kahneman and Tversky's prospect theory. The combined disappointment and weighting functions result in valuations of MBGs that are different from those implied by using the expected net benefit criterion, both because of the extra valuation given to potential disappointment from a bad decision, as well as the weighting of probabilities that gives bigger weights than merited to small likelihood outcomes. Because return probabilities are relatively small, consumers may overweight them, and that may lead to higher premiums for MBGs relative to the standard expected net benefits model. In interpreting the empirical results, we compare the estimated premiums for MBGs when consumers use their actual return probabilities to assess MBG contracts with the estimated premiums when we assume weighted probabilities, and see how the weighting and disappointment together increase the "expected" cost of a loss should the product not be a fit.

In the next section we develop and present the extended prospect theory model. In Section 3 we present the survey design, and in Section 4 we estimate behavior with micro-level data to calibrate empirically what the behavioral parameters would need to be in order to use disappointment aversion as an explanation. In this context, we estimate the perceived value of MBGs to consumers and provide an explanation for how consumers generate these MBG valuations. In particular, we test whether prospect theory is a relevant framework for valuing the MBG. We conclude with a discussion of the implications of our findings.

2. Modeling consumes choice

2.1. The value and properties of MBG contract

In this section we build a simple model, which is based on expected utility principles, and show that the basic properties of

MBG contracts can be explained by using a simplified model that is consistent with rational choice. A consumer considers purchasing a product with uncertainty regarding the benefits of using the product. Previous studies have modeled uncertainty about benefits, i.e., consumers are aware to the distribution of benefits but do not know their idiosyncratic location (Hahn, 2005), uncertainty about fit (Heiman and Ofir, 2010) or about both fit and benefit (Heiman et al., 2002; Nizovtsev and Novshek, 2004). In Section 2.2 we analyze MBGs testing whether loss aversion (Kahneman and Tversky, 1979) and disappointment aversion (Gul, 1991) combined with a multiple reference point can better explain consumers choice of MBG contracts.

In order to be consistent with this literature, which demonstrated or developed its theory on consumers' choices between lotteries, we assume that consumers are uncertain about whether or not the product fits their needs. Let V denote the benefit of a random consumer from a product if it fits consumer's needs and expectations and zero benefit otherwise. The purchase of a product without MBG can be seen as a lottery denoted by two pairs of value and probabilities. The probabilities of fit and misfit are denoted by q and $1-q$ respectively. The price of a product sold without MBG, is denoted by $P_{t=0}$ (the subscript $t=0$ indicates that the duration of the MBG is zero). We assume that the price paid is rather small relative to reference wealth, W allowing us to assume a linear utility function similar to Kőszegi and Rabin (2007). The expected utility of a random consumer without MBG is given by: $E(U_{t=0}) = q(V - P_{t=0}) - (1-q)P_{t=0} + W$. A consumer will purchase the product if $E(U_{t=0}) \geq 0$. Thus all consumers will purchase the product when $V \geq \frac{P_{t=0}}{q}$. Now suppose that the product comes with a MBG with a duration of t days. Given the baseline duration $t = t_0$ consumers experience the product and learn about their idiosyncratic probability of fit.

Learning calibrates consumers' initial predictions about the probability of fit that is based upon prior experience and indirect information, i.e., advertisements, or other consumers' beliefs and sellers' information. Studies on the effect of learning on decisions under uncertainty have shared the assumption that consumers maximize expected utility given the information they possess at the time of decision making. New information about the product, which is received before the decision is taken, is integrate with prior beliefs using the Bayesian updating rule (Erdem and Keane, 1996; Meyer and Sathi, 1985; Roberts and Urban, 1988). Posterior benefits are assumed to be drawn from a distribution such as a uniform or log-normal (Roberts and Urban, 1988). Thus an increase in the stock of information resulting from buying a longer MBG contract is expected to reduce uncertainty. The lower uncertainty does not imply that the longer the MBG the higher the likelihood of purchase, however. The probability of purchase depends on the distributions of priors and the sampling signal (Halbheer et al., 2012). Arrow's (1962) seminal paper expressed learning as a concave function of experience, a finding that since then has been supported by the management economics literature (e.g., Day and Montgomery, 1983; Lieberman, 1987), as well as by the psychology literature, where the "power law of practice" implies that learning is concave with respect to trials (Delaney et al., 1998; Newell and Rosenbloom, 1981). Following the aforementioned studies we assume that the learning curve $L(t)$ is concave with respect to the duration of the MBG contract, i.e., $L'(t) > 0$ while the change in marginal learning can be negative $L''(t) < 0$. $L''(t)$ changes sign when learning is assumed to follow a sigmoid learning pattern (Parente, 1994; Young, 2009). Let $f(t) = \frac{1}{L(t)} \int_0^t L(z)dz$ where $f(t=0) = F_0 = 0$ and $f(t=T) = 1$ denote the accumulated learning during a MBG contract with the duration of t days. After trying the product for t days a buyer discovers whether the product fits with probability of $f(t)$. The probability of discovering a poor fit is thus given by $(1-q)f(t)$. A random consumer will return the product if a misfit is realized and the cost of returning the product is smaller than the benefit of keeping a product that does not fit, i.e., $P_t - R > 0$. If the later condition is not satisfied than the MBG contract is worthless. A consumer will not discover whether or not the products fit with probability of $1-f(t)$. The joint probability of not discovering a misfit and that the product will yield low benefit is thus given by $(1-q)(1-f(t))$. The expected utility of a random buyer who purchases with MBG contract with the duration of t_0 is given by:

$$E(U_{t_0}) = q(V - P_{t_0}) - (1-q)f(t_0)R - (1-q)(1-f(t_0))P_{t_0} + W \quad (1)$$

A consumer will purchase the product if $E(U_{t_0}) \geq 0$. The value of the MBG option with the duration of t_0 equals to the difference between the expected value with and without MBG is represented by:

$$E(V_{t_0}) - E(V_{t=0}) = (1-q)f(t_0)(P_{t_0} - R) + P_{t=0} - P_{t_0} \quad (2)$$

The threshold price difference that makes a risk neutral consumer indifferent between buying with and without an MBG (if both alternatives are offered at the same retail establishment simultaneously) is represented by:

$$\Delta P_{t=0}^{t_0} = P_{t_0} - P_{t=0} = (1-q)f(t_0)(P_{t_0} - R). \quad (3)$$

Consumers' willingness to pay for an MBG of duration t_0, i.e., $\Delta P_{t=0}^{t_0} = P_{t_0} - P_{t=0}$, increases with the probability of discovering that the products does not fit, i.e., $(1-q)f(t_0)$, as well as with product price, and declines with return cost R.

Let $\mathrm{RP} = (1-q)f(t_0)$ represent the return probability and $\mathrm{RP}' = (1-q)f'(t_0)$ therefore represents change in return probability as the duration of the MBG is changed. Now suppose that a random consumer considers a change in the duration of the MBG contract. For the consumer to be indifferent between the two MBG durations, the expected payment for the change in the duration of a MBG contract must be equal to the marginal change in the expected benefits from this change in duration and it is given by:

$$(1-q)\frac{\partial f(t)}{\partial t}(P_t - R) - (1 - (1-q)f(t))\frac{dP_t}{dt} = \mathrm{RP}'_{t_0}(P_t - R) - (1 - \mathrm{RP}_{t_0})\frac{dP_t}{dt} \quad (4)$$

The first element represents the increase in the "insurance value", i.e., lowering the cost of being stuck with a product that does not fit, while the second element represents the loss from purchasing a better insurance contract when the contract is not exercised. The change in price with respect to changes in the duration of the MBG contract for which the buyer will be indifferent is thus given by:

$$\frac{dP_t}{dt} = \frac{(1-q)\frac{\partial f(t_0)}{\partial t}}{1-(1-q)f(t_0)}(P_t - R) = \frac{\mathrm{RP}'(t)}{1-\mathrm{RP}_t}(P_t - R) \quad (5)$$

It is easy to see that $\frac{dP_t}{dt} > 0$ and Hypothesis (1) is thus offered.

Hypothesis 1. Increases in the duration of the return period increase the premium consumers are willing to pay for a product with an MBG, i.e., $\frac{\partial P_t}{\partial t} \geq 0$.

The sign of the change in the marginal WTP depends on the specific form of the learning curve. Specifically if the learning curve is S shaped than the sign of the change in the marginal WTP may be non-negative in earlier periods when marginal learning is very high[1] other than that the second derivative of price with respect to time is negative throughout the possible durations of the MBG contract due to declining marginal learning. We assume that the second derivative of learning. with respect to time is negative throughout the possible durations of the MBG. This implies the following hypothesis

Hypothesis 2. Increases in the duration of the return period lead to smaller changes in consumer willingness to pay for these additional durations, i.e., $\frac{\partial^2 P_t}{\partial t^2} < 0$.

[1] The sign of the second derivative of WTP is negative if $\frac{2(1-q)(f')^2}{(1-RP)} < -f''$.

Total differentiation of Eq. (5) indicates that consumers will demand a higher price discount for buying without MBG when the probability of returns RP is higher, the product is more expensive and the cost of return is lower, i.e.,

$$\frac{dP_t}{d\mathrm{RP}} > 0;\ \frac{dP_t}{dP_{t=0}} > 0;\ \frac{dP_t}{dR} < 0.$$

These results enable us to hypothesize the valence of the signs resulting from changes in MBG's quality and importance. We study differences in consumer valuation between MBG on store and online (catalog) purchases and that between money and store credit refunds.

Consumers return cost is a crucial variable that determines the value of MBG and the derived price with MBG. Previous studies suggested that increasing the return cost is an efficient tactic in controlling returns (Chu, Gerstner, and Hess, 1998; Davis, Hagerty, and Gerstner, 1998) other scholars suggested that the polar opposite strategy of offering unconditional and generous guarantees is more profitable (Petersen and Kumar, 2009; Petersen and Kumar, 2010; Reinartz and Kumar, 2003). One way to evaluate the relative importance of the weight of return cost is to estimate its marginal contribution to price using cross retailer's offers of the same product category (Posselt, Gerstner, and Radic, 2008). However, this procedure may produce noisy results due to variation in products, retail services and retailers' cost of handling. Another way to proceed, as we propose in this paper, is to estimate the difference between the value of MBG and store credit at the same hypothetical retail establishment.

Let $P_{t_0}^{SC}$ denote the price consumers will be willing to pay for the purchase of a good with a t_0 return period where the consumers receive store credit rather than cash upon returning the product. Since the return cost of store credit are higher than that of cash it is hypothesized that

Hypothesis 3. (store credit < cash). Consumers' value store credit returns less than cash returns. (We provide a formal proof in Appendix A1.)

We now employ the finding that $\frac{dP_t}{d\mathrm{RP}} > 0$ in examining the differences in willingness to pay for MBG between online (catalog) and brick and mortar stores that offer the same product. The probability of fit, q is lower for remote purchases since buyers are unable to inspect and test products before purchase (Akaah and Korgaonkar, 1988; Ofek, Katona and Sarvary, 2010; Van den Poel and Leunis, 1999) and because of the time lag between ordering and receiving the product, the consumer's perception of the product may be distorted (Wood, 2001). Data collected by the national federation of retailers (NFR) in the U.S. indicate that overall customer return rates are about 8.7% of retail sales. Return rates are significantly higher for catalog and e-commerce retailers, where Internet purchases are about twice as likely to be returned as are brick-and-mortar retail purchases,[2] and catalog purchase return rates are similarly estimated to be around 16% (Petersen and Kumar, 2010). Consistent with this evidence we assume that the probability of fit for catalog purchases is smaller, resulting in lower probability of fit, i.e., $1-q^S < 1-q^C$, and higher return probability for catalog purchases relative to store purchases, i.e., $\mathrm{RP}_t \le \mathrm{RP}ca_t$, where the S and C superscripts denote store and catalog (online) purchases respectively. Lower net duration of online MBG, which may result from countdown of the MBG period from the day of shipment and not the day of arrival reduce $f(t_0)$ and that may counterbalance the lower fit. The majority of retailers who operate dual online and brick and mortar channels offer identical MBG contracts in both cannels,[3] others such as Amazon compensated the online buyer for the loss of experiencing time due to shipping time by starting the day count from the day the product arrived to the customer.[4] If the retailers offer identical length of MBG contract to online then unless the time constrain is not binding, $f(t_0^C) < f(t_0^S)$. The lower probability of fit and the lower learning intensity affect the probability of returns in opposite directions, the former increases it and the later decreases it. The effect of online (catalog) lower probability of fit dominates the learning effect if $\frac{1-q^C}{1-q^S} > \frac{f(t_0^S)}{f(t_0^C)}$. For the standard duration of MBG and shipment given reasonable speed of learning the probability of returns for products that are bought online is higher than that of brick and mortar purchases, i.e., $(1-q^C)f(t_0^C) > (1-q^S)f(t_0^S)$. This results in higher value of MBG for online (catalog) purchases relative to brick and mortar stores. Hypothesis 4 is thus offered:

Hypothesis 4.a. MBGs for online and catalog purchases are valued more than are MBGs for brick-and-mortar purchase.

Hypothesis 4.b. Additional durations of MBG are valued more for Internet purchases than for brick-and-mortar purchases.

Consumer heterogeneity with respect to the probability of product return affects the value of MBG (Heiman et al., 2002; Nizovtsev and Novshek, 2004). Consumers whose probability of fit are lower or who learn more through the MBG are more likely to return products more frequently. The relevant factor in determining the value of MBG is the probability of return, or $(1-q)f(t_0)$, and not just the probability that the product does not match, i.e., $1-q$, as has been assumed in previous studies that ignore learning effects. Just as we argued that MBGs will have a higher valuation for Internet purchases because of the higher likelihood of product return, we can argue that consumers that return products more will have a higher valuation of MBGs, and similarly will have a higher valuation of marginal extensions in the MBG duration, i.e., $\frac{dP_t}{d\mathrm{RP}} > 0$ and Hypothesis 5 is therefore offered:

Hypothesis 5.a. Consumers with a higher than average return probability will assign a higher valuation to MBGs.

Hypothesis 5.b. Consumers with a higher than average return probability will assign a higher valuation to incremental duration of MBGs.

Changes in the duration of MBG affect both the probability of returns and the price of the product. These changes can be viewed as creation of a new lottery with higher probability of a bad outcome (in the case of lower duration) and higher price. The expected utility framework, which was presented earlier, suggests that consumers would choose the alternative with the highest expected benefit. Suppose that a retailer offers two MBG contracts one of 30 days and the other of 7 days for a discount of $1-\beta$%. A consumer who would have chosen to purchase a product with 30 days MBG now needs to decide whether to shift to a 7 days MBG contract and buy with a reduced price. A consumer is expected to prefer the 7 day MBG over the standard 30 days if $(1-\beta)P_{30} \ge -(1-q)(f_{30}-\beta f_7)P_{30} - (1-q)(f_{30}-f_7)R$. This condition asserts that by choosing a 7 days MBG contract the consumer gains $(1-\beta)P_{30}$ but this gain is expected to be counterbalanced by the increase in the expected loss $(1-q)(f_{30}-\beta f_7)P_{30}$ (lower probability of returns). However there are many empirical studies that showed violation of this prediction suggesting that the expected loss is evaluated as larger than an equivalent monetary gain (Kahneman and Tversky, 1979) or that consumers make choices based on multiple reference points (Kőszegi and Rabin, 2007, 2009) that are not necessarily zero as implied by (Kahneman and Tversky, 1979). Choosing between MBG contracts is definitely a complex task since it refers to three outcomes: the product fits, the product does not fit and the consumer succeeded in discovering it when the product could be returned

[2] http://www.skipmcgrath.com/articles/ebay-secondary-surplus-liquidation-market.shtml

[3] For example The GAP states that returns must be made "Within 45 days of purchase."

[4] "Amazon.com, and many sellers on Amazon.com, offer returns for most items within 30 days of receipt of shipment"

and the discovering a poor match too late and owning a product that provides low net value. In addition consumers may refer to the norm of 30 days if they considered buying the product. Or do they refer instead to the reference point of not buying at all? In Section 2.2 we extend and apply the prospect theory to model consumers' choices of MBG with varying duration.

2.2. The extended prospect theory approach

As an alternative to the rational EU framework, we analyze consumers' valuation of alternative MBG arrangements within a modeling framework that builds on Kahneman and Tversky's (1979) prospect theory. Various researchers have built on their findings, with a common assumption being that when consumers experience a loss with respect to their reference position, they experience an additional psychological cost that reflects the consumer's aversion to loss. While prospect theory has been widely discussed, there continues to be debate about how reference points are selected and employed in decision-making. Gul (1991) supposes that outcomes of a gamble are compared to the certainty equivalent (CE). Others, suppose that the outcomes are measured against the (reference based) expected utility (Shalev, 2000); or, against the best forgone alternative (Loomes and Sugden 1982). Kőszegi and Rabin (2006, 2007, 2009) provide some evidence that when facing a choice with many dimensions, each dimension is measured against its own reference point. They model this as a sum of referenced based value functions.

In the case of a product purchases, we can consider the reference state to be the consumer's anticipation of full benefits received from the consumption of the purchased good. If the consumer happens to realize that the purchased product does not, in fact, fit her needs, she is disappointed. Specifically, "disappointment" in this case reflects the psychological loss, above and beyond the price paid for the product, for having made a bad purchase decision in which the expected benefits can no longer be realized. We incorporate psychological disappointment as an additively (negative) separable fixed cost in the model, a practice that is consistent with the literature (e.g., Bell, 1982, 1985; Loomes and Sugden, 1982, 1986; Kőszegi and Rabin, 2007, 2009).[5]

In our model a consumer who considers buying an MBG with a given duration weights three possible outcomes before deciding: (1) *satisfied*, i.e., the product fits; (2) *returned*, i.e., the buyer discovers that the product does not fit her needs within the guaranteed period and returns it for a full refund; and (3) *disappointed,* i.e., the buyer discovers that the product does not fit her needs, but only after the MBG expires. The probability of realizing each of these three states depends not only on the duration of MBG, but also on the choice between retail channels, where buying online may save money but is riskier, and some retail outlets may offer only store credit rather than a full cash refund. Thus integration of prospect theory principles and those of multiple reference points should add value to the willingness to pay that was calculated according to expected utility and presented in Hypotheses 2–4.

A consumer who consider buying the shorter (longer) MBG is taking in mind that the shorter duration decreases (increases) the probability of discovery of non-fit increasing the likelihood of disappointment on one hand while benefiting from paying lower price that increases the net value if satisfied and lowering the likelihood of the state "returned" on the other hand.

We combine three elements from prospect theory, reference point, concavity in gains and convexity in losses; and a probability weighting function with disappointment aversion, and choice under risk considering reference points in separable functions within the utility specification. In particular, the weighted benefit given a specific MBG duration is evaluated with respect to a fixed, default return period, which we denote t_0. Different evaluations of gains and losses are captured by assigning a valuation function to each possible outcome, with greater weight given to losses than to gains with respect to the reference point. In our case, this will be achieved by assigning a "disappointment" loss to the negative outcome. The third element, which is integrated, the probability weighting function $\pi(\cdot) : [0, 1] \rightarrow [0, 1]$, represents the systematic distortion of the probabilities they perceive.

While the probability weights are not subjective probabilities per se, individuals behave as if the value of the probability weights were the true probabilities. Hence, for convenience of discussing the effects of probability weighting, we will refer to $\pi(p)$ as the perceived probability, which is a function of the true probability. According to Kahneman and Tversky (1979), the perceived probability increases with the actual probability, small probabilities are overestimated, and large probabilities are underestimated. Thus, the probability of a non-fit, $1-q$, translates into a perceived probability of non-fit $\pi(1-q)$, which is decreasing in q: $\pi'(1-q) < 0$. And, for some given level of expected fit $\bar{q}$, a $\hat{q}$ larger than $\bar{q}$ implies $\pi(1-\hat{q}) > 1-\hat{q}$ and $\pi''(1-\hat{q}) < 0$, while a $\tilde{q}$ less than $\bar{q}$ implies $\pi(1-\tilde{q}) < 1-\tilde{q}$ and $\pi''(1-\tilde{q}) > 0$.

To assess the value of a change in the MBG duration, we set the duration of a baseline MBG arrangement as 30 days, i.e., $t_0 = 30$ days,[6] which is selected as the reference point in the empirical application because of evidence that this is the most common duration for MBGs in the United States (Heiman et al., 2002) and what represents the base (reference) case in our questionnaire. Let ΔP_t denotes the change in the product price and $\Delta f(t)$ denotes the change in learning associated with a change in the MBG duration from t_0 to t. A longer return period ($t > t_0$) is likely to result in a higher return probability from learning $(1-q)\Delta f(t) > 0$ as well as a higher price ($\Delta P_t > 0$). The higher return probability reduces the probability that the consumer will be disappointed by being stuck with a product that does not fit her needs.

While the outcomes in standard prospect theory models are unidimensional (generally monetary payoffs), the outcomes in our model are multidimensional. Tversky and Kahneman (1991) generalized prospect theory to allow for multiple dimensions, with reference points measured in product space. Heidhues and Kőszegi (2008) develop a model in which consumers refer to two reference points: the expected price of a product and its distance from idiosyncratic preference, i.e., fit. Our approach is similar, though our reference points are measured in monetary outcomes and the duration of MBG.

Following Kőszegi and Rabin (2007) who suggested that when the choice involves relatively small expenditure relative to income we define the valuation function (the equivalent of the utility function) in monetary terms. Under this assumption the monetary loss has the same weight as the gain, whereas the notion of loss aversion as presented in prospect theory suggests concavity in gains and convexity on losses. Concavity or convexity is captured in our model through $f(t)$ that yields non linear valuation functions.

Integrating the notion of the consumer's probability weighting function, π, into Eq. (5), and recognizing that the consumer now gains a net value of $(P_{t_0} + D - R)$ when returning a product under MBG (i.e., she eliminates her disappointment), implies that the consumer's willingness to pay for a marginal change in duration can be

[5] The literature generally treats the psychological loss or gains component as a function of the size of the loss or gain of the realized outcome relative to the expected reference outcome. For MBGs the only relevant alternative state is when the consumer realizes a loss (product non-fit), so we only consider disappointment (loss). Moreover, while we would expect consumer disappointment to depend on the size of the purchase, our price is fixed, so we adopt a fixed disappointment level.

[6] Neither the theoretical nor empirical results depend on the choice of reference duration since loss aversion is already captured separately through the disappointment parameter.

expressed as:

$$\Delta P_t = \frac{\pi\left((1-q)\frac{\partial f(t)}{\partial t}\right)(P_{t_0}+D-R)}{\pi(1-(1-q)f(t))} = \frac{\pi'(\mathrm{RP}_t)}{\pi(1-\mathrm{RP}_t)}(P_{t_0}+D-R) \tag{6}$$

where, in the new notation given in the right term, RP represents the probability of returning the product and $\Delta\pi(\mathrm{RP})$ represents the change in perceived probability of returning the product induced by the change in MBG duration. Eq. (6) suggests that the maximal (minimal) amount that consumers will be willing to pay (receive) for longer (shorter) MBGs is a product of the increase (reduction) in the RP and the gain from returning an unfit product, divided by the probability of keeping the product. Note that Eq. (6) is an extension of Eq. (5) which denotes the willingness to pay for longer MBG under EU. The two additional elements in (6) are the weighting function (π) and the addition of the disappointment cost D.

Consumers learn over time whether the product is a fit or not as they experience the product. Marginal learning with respect to time is reflected in the equations by $\Delta\mathrm{RP}_t$, which is adjusted by the probability weighting function in the form $\pi(\Delta\mathrm{RP}_t)$. We have assumed that learning is concave with respect to the number of trials, and that product trials are a linear function of time (for example, the consumer uses the shirt once a week), leading us to conclude that learning about fit is a concave function of time: $\mathrm{RP}'_t > 0$; $\mathrm{RP}''_t < 0$. Following Kahneman and Tversky (1979), it is assumed that the probability weighting function itself, $\pi(\cdot)$, is concave in the area of small probabilities, so $\pi(\Delta\mathrm{RP}_t)$ is concave. The combined concavity of both the learning and the probability weighting functions implies that the perceived probability of return increases with the duration of the return period, while this marginal effect declines, i.e., $\partial\Delta\pi(\mathrm{RP}_t)/\partial t > 0$ and $\partial^2\Delta\pi(\mathrm{RP}_t)/\partial t^2 < 0$, independently of whether t is greater than or less than t_0, although $\Delta\pi(\mathrm{RP}_t) < 0$ if $t < t_0$, and $\Delta\pi(\mathrm{RP}_t) > 0$ if $t > t_0$.

To illustrate the order of magnitude of a change in price, let us consider the case where the reference price of the product is \$40 and the reference return period is 30 days, as in the case of our survey. Now suppose the perceived probability of return $\pi(\mathrm{RP}_{t_0})$ under this initial arrangement is 10%. If the return cost is \$5.00 and the disappointment cost is \$10 then the minimum discount a consumer will demand for buying without the MBG at all is \$5.00 (i.e., $0.1 \times \$45/0.9$, where $\$45 = P + D - R$). Now consider the case in which the consumer wants to extend the MBG to a longer return period, say, 60 days, and assume that this will increase the perceived return probability, $rm\Delta\pi(\mathrm{RP}_t)$, by 5%. The updated perceived probability of retaining the item under the longer period, $\pi(1-\mathrm{RP}_{t=60})$, is therefore 85. The extra amount the customer would be willing to pay to extend the duration of the MBG to 60 days is \$2.65 (i.e., $.05 \times \$45/.85 = \$2.65$). Changes in D and the weighting function can significantly alter the premiums consumers are willing to pay for MBGs of different durations. For example, if $D = \$5$, then the additional amount the consumer will pay for the 60-day MBG declines to \$2.35. One of the main purposes of our empirical analysis is to estimate the order of magnitude of disappointment using the weighting function that where estimated in Tversky and Kahneman (1992).

The value of the MBG to the customer is much lower when disappointment is not accounted for. The expected value of an MBG would otherwise be \$3.89 (when $D = 0$), and the value of extending the MBG to 60 days would be \$2.05. Existing evidence, such as the empirical finding that MBGs increase demand by 19% (Anderson et al., 2009). The value of MBG and changes in its duration are lower than implied by the integrated prospect and disappointment theories. The expected value of MBG as presented in Eq. (3) given these figures is \$3.5 and the value of additional 30 days is \$1.75 less than implied by Eq. (5) and lower than that implied by empirical findings that suggest that MBG may increase demand by 19% (Anderson et al., 2009). Hypothesis 6 is thus suggested

Hypothesis 6. Consumers will have a greater willingness to pay for MBGs than is implied under expected benefit analysis. That is, $D > 0$.

While learning and the probability function assures concavity, our assumption that the disappointment function is linear is different than that presented in previous studies (Gul, 1991; Kőszegi and Rabin, 2007, 2009). We expect that the combination of weighting probabilities and disappointment component of utility will provide a better fit relative to weighting probabilities or disappointment alone.

In order to test the six hypotheses presented in the previous section, we use data from a survey that was conducted in northern California. In the next section we describe the design of the survey, questionnaire design and the characteristics of the sample.

3. Methodology

3.1. Design of data collection

We used a hypothetical survey with real shoppers in order to estimate consumers' willingness to pay a premium for extending, and accept discounts for reducing, the MBG duration. The limitations of hypothetical surveys are well documented. However it provides several advantages that allow us to isolate the demand side (consumers) from the supply side. Estimating consumers' willingness to pay (accept) from real market data taken from a retailer (or retailers) without having information on the demand side for returns (the retailer's MBG policy) is theoretically problematic and may lead to misleading results since the returns recorded reflect equilibrium rather than consumers' valuation (see Nerlove, 1995, p. 1715). Market data provides valid estimations only if a retailer's policy is exogenous of products' returns, which is an extreme assumption given the findings that retailers influence returns by controlling the cost of returns through hassle and restocking fees (Hess, Chu and Gerstner, 1996; Posselt, Gerstner and Radic, 2008). Furthermore, the findings that the relationship between the firm's profitability and the rate of product returns is an inverted-U shape (Reinartz and Kumar, 2003; Venkatesan and Kumar, 2004) indicates that firms control returns either by better selecting the products, suppliers, and/or customers, and calibrate their return policies. A laboratory experiment has the benefit that consumers can select between alternatives and their choices can be analyzed through revealed preferences, yet it has its shortcomings such as an artificial environment, and student responders. This work extends and applies prospect theory which has history of testing based on survey data (Thaler, 1980). Hypothetical survey can provide reasonable predictions when changes in marginal WTP and WTA are to be estimated (Louviere, Hensher and Swait, 2000; Lusk and Schroeder, 2004). It is therefore natural to use survey technique, and the decision to conduct the survey in shopping malls has the advantage of interviewing shoppers in a relevant place.

Thus the variables that need to be estimated are past experience with returns, the willingness to pay for a baseline and extended/shortened MBG durations, and disappointment net of return costs. Identifying the proportion of products returned in the past captures the combination between making a wrong choice and the probability of identify that the product is "wrong" during the duration of the MBG. The estimation of the willingness to pay for the MBG option with respect to changes in duration represents the change in the probability to discovering a "wrong" choice as duration increases (decreases).

3.2. Questionnaire design and procedure

The questionnaire consisted of 10 questions: seven were open-ended questions and three are close-ended questions referring to socio economic variables. The open questions asked consumers about

their share of purchases in stores and catalogs, items that are returned more frequently, their willingness to accept a shorter duration MBG given a reference of 30 days or to get the refund in credit, and their willingness to pay for a longer (60-day) MBG. For example:

How much off on this $40 item would you demand if:

(a) it did not come with a MBG? $_____;
(b) it came with only a 7-day MBG? $_____;
(c) if you only received store credit or replacement instead of your money back? $_____

Respondents were randomly chosen at various shopping malls and instructed to assess their personal value of an item of clothing purchased with varying levels and types of MBGs. On average, the survey took about 10 min to complete. The survey design (ordered questions highlighting the duration changes) has the potential to produce a demand effect. However, given the strong norm in the US of a 30 day MBG, we felt the effect of framing in reference to the 30 day MBG would be detectable and important. This demand effect is typical of the reference effect that would occur in the real world with non-standard MBGs. Moreover, we sought to make the questions and mechanism as transparent as possible to create internally consistent answers in accordance with the principles of the non-market valuation literature. While this does create the possibility for bias, recent work suggests that the bias should be relatively mild relative to alternatives such as conjoint analysis (Miller et al., 2011).

3.3. Sample

Of the 222 consumers interviewed, 48% were male and 52% were female. The average age of respondents was 27, where 16 were below the age of 18. The oldest respondent was 76 years old. Thirty-four responses were rejected, primarily because they failed in providing information that is consistent with rational decision making, e,g., they demanded more than the price of the product for shorter MBG, leaving 188 usable responses.

4. Empirical estimation

4.1. The effect of changes in MBG's duration on MBG value

The baseline MBG option was a 30-day MBG on a $40 item of clothing (a shirt). Respondents were asked the discount required for them to accept not having an MBG and to get a 7-day MBG on the same article of clothing, as well as the monetary compensation required to switch from a 30-day full cash refund guarantee to a 30-day store credit guarantee (SCG). They were also asked for their willingness to pay a premium to extend the MBG to a 60-day and to an infinite-length MBG. These same questions were repeated for the same item ordered from a catalog. Individuals were asked to estimate the percentage of clothing purchases they returned (both in store and through catalogs) during the last year.

On average, respondents demanded a $10.43 discount to forego the 30-day guarantee and buy the $40 shirt without guarantee. At 25% of the original product price, this assessment seems high given the reported 10% probability of a return. However, there is anecdotal evidence leading us to believe that individuals are actually willing to pay such high prices for guarantees. Heiman et al. (2002) provide some evidence that buyers are willing to pay between 5 and 30% extra for return options in various product categories. The percentage of clothing items returned in stores (10.13% on average) and through catalogs (18.63% on average) by our sample is similar to that reported in the professional literature cited earlier of 8.7% for regular store sales versus 16% for catalog sales.

Fig. 1 displays the heterogeneity in average RP within our sample. While over half of the respondents return 10% or less of clothing purchases, a significant number return more than 20% of these purchases. Similarly, Table 1 reveals a great deal of heterogeneity in the discount required for consumers to accept not having any MBG. For in-store purchases, almost three-tenths of the sample requires a discount of under $5.00, while over a third requires over $15. The average discount required is $10.43, and the median is $10. The modal response was $0, and a $10 discount was the second most frequent response. Responses were highly variable with nearly every whole number between $0 and $15 represented. For catalog purchases, the results were similar, although the number of respondents expressing larger discounts was higher, with a mean of $11.84.

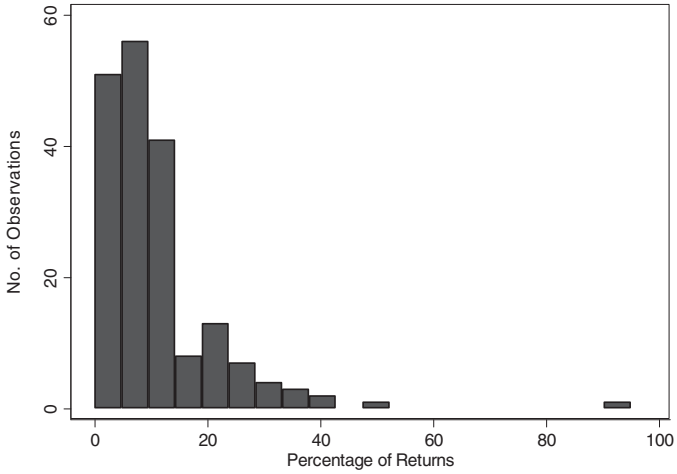


Fig. 1. Distribution of average returns.

Table 1
Discounts necessary to accept not having an MBG for in-store and catalog purchases.

Discounts	30-day MBG for in-store purchases		30-day MBG for catalog purchases	
	Number	%	Number	%
Under $5	54	29	59	31
$5 to under $10	28	15	17	9
$10 to under $15	41	22	28	15
$15 to under $25	48	26	58	31
$25 to $40	17	9	26	14
Total	188	100	188	100
Mean	$10.43		$11.84	
Median	$10.00		$10.00	

The results that are presented in Table 1 support Hypothesis 6 that consumers will have a greater willingness to pay for MBGs than is implied by expected benefit analysis. Respondents reported, on average, about a 10% return rate, which, given that they pay $40 for the product suggests that they should value the MBG at most at $4, and even less if they account for the hassle costs of returning the product.

Because Hypotheses 1 through 4 are not conditional on the parameters of the model, it is possible to test these hypotheses without estimating Eq. (4). While we appeal to the model to derive our hypotheses, the hypotheses are stated in terms of comparing valuations over time, so our tests will be based on these comparisons.

For each individual, we recorded changes in discounts/premiums for ΔMBG^{0}_{30}, ΔMBG^{7}_{30}, ΔMBG^{60}_{30}, ΔMBG^{∞}_{30}, and $\Delta MBG^{t,catalog}_{t,store}$ for both in-store and catalog purchases, where the comparison group is the 30-day MBG (hence, $\Delta P^{30}_{30} = 0$). This information allows us to construct data on the slope of each segment. We can then check directly for violations of the four hypotheses, and use classical hypothesis testing to check for significant violations. Since a precursory examination of the data support Hypothesis 2 (that the marginal value of MBG duration declines as the MBG duration increases) over the full range of observations, the following tests will be based on a decreasing marginal

Table 2
Violations of hypotheses.

Hypothesis	Violations	Possible violations	Individuals violating	Results of regression tests
Hypothesis 1	7	1422	5	Reject violations, $p < 0.01$
Hypothesis 2	85	1046	51	Reject violations, $p < 0.01$
Hypothesis 3	0	361	0	Reject violations, $p < 0.01$
Hypothesis 4	56	698	34	Fail to reject violations[a]
All hypotheses	148	3527	66	

The results are based on a total of 181 respondents.
[a] Can reject violations for one pair of line segments at the 0.01; cannot reject at any reasonable level for the other three pairs.

value of MBG duration everywhere. In the spirit of Camerer's (1989) tests of risk and uncertainty models, we examine the primary testable hypotheses of our model using individual data. Table 2 displays the number of violations and the number of possible violations for each hypothesis. While the table shows that a significant number of individuals (36%) violate the hypotheses at least once, the violations occur only 4% of the time. It would be hard to reject the notion that these violations were occurring due to idiosyncratic errors.

We can test Hypotheses 1 through 4 by jointly testing all restrictions included in each hypothesis for rejection. However, this test holds little power and, indeed, cannot be rejected for any of our hypotheses. A more powerful method is to independently test for each possible failure, maintaining failure as the null hypothesis. The results of these hypotheses appear in the far right column of Table 2. These results are highly consistent with our model and hypotheses. Only Hypothesis 4, which suggests that consumers are expected to have a higher valuation of marginal changes in MBG duration for catalogs than for store, fails to reject violations in some cases. Even so, the estimates reflect congruence with Hypothesis 4.

By regressing the slope parameters on a constant, it is possible to test these hypotheses using traditional hypothesis testing. We employ a regression structure rather than a direct means test to take advantage of the correlated error structure from survey data. Hence, we employ Zellner's Seemingly Unrelated Regression (SUR) technique to estimate the means. The results of this estimation appear in Table 3.

The results in Table 3 clearly support Hypotheses 1, 2 and 3. However, Hypothesis 4, that consumers value additional MBG durations for catalog purchases more than for store purchases, is statistically supported only when extending the duration from 7 days to 30 days. For other periods, the value of extending the duration of catalog versus store MBGs are not significantly different. The lack of a significant difference in the value of MBG for the first 7 days may, as mentioned earlier, relate to the delay in the consumer receiving the product purchased by catalog, since the consumer cannot get any value from having an MBG until the product is received and tested. The significant difference in valuation of MBGs between the 7th and 30th days of the guarantee can be attributed to the excess uncertainty from not being able to test a catalog product before purchase. After 30 days, the differences in marginal valuations of MBGs are no longer significant, which suggests that the marginal rates of learning are no longer significantly different after one month.

Table 3
Calculated per-day value of changes in MBG duration and conditions.

Variable	Mean ($)	Standard error ($)
$\Delta P^{7}_{0,\text{store}}$/7 MBG/day 1–7 store	0.68	0.06
$\Delta P^{30}_{7,\text{store}}$/23 MBG/day 7–30 store	0.25	0.02
$\Delta P^{60}_{30,\text{store}}$/30 MBG/day 30–60 store	0.07	0.02
$\Delta P^{\infty}_{60,\text{store}}$[a] store	2.62	0.34
$\Delta P^{30.\text{SC}}_{30,\text{cash}}$ store	−6.78	0.76
$\Delta P^{7}_{0,\text{cat}}$/7 MBG/day 1–7 catalog	0.73	0.07
$\Delta P^{30}_{7,\text{cat}}$/23 MBG/day 7–30 catalog	0.35	0.03
$\Delta P^{60}_{30,\text{cat}}$ MBG/day 30–60 catalog	0.08	0.02
$\Delta P^{\infty}_{60,\text{cat}}$[a] catalog	2.76	0.41
$\Delta P^{30,\text{CSC}}_{30,\text{cash}} \leq 0$ catalog	−8.95	0.79

[a] Because a slope cannot be calculated, we estimate the average rise of $\Delta\text{MBG}^{\infty}_{60}$.

4.2. Heterogeneity and disappointment aversion

Testing Hypothesis 5 requires estimation of the relationship between the return probability and the premiums/discounts for various-length MBGs. The estimation in the previous section (Table 3) determined the average value of an additional day of MBG over different lengths of MBGs ($0.67 per day for the first 7 days, $0.25 for the next 23 days, and $.07 for the next 30 days). In this section we examine whether buyers with different return probabilities have differing valuations. We approximate the relationship in Eq. (6) using a quadratic form for the relationship between the return probability and the willingness to pay (receive) for extending (shortening) the MBG duration:

We approximate the relationship in Eqs. (5) and (6) using the form (7)

$$\Delta P^{t}_{30} = \sum_{t=0,\,7,\,60,\,\infty} (\beta_t + \beta_{\text{RP}t}\overline{\text{RP}})\delta_t + (\beta_{\text{SC}} + \beta_{\text{SC,RP}}\overline{\text{RP}})\delta_{\text{SC}} + \beta_{+}\overline{\text{RP}}^{2}\delta_{t>30} + \beta_{-}\overline{\text{RP}}^{2}\delta_{t<30} \quad (7)$$

where δ_t is a dummy variable taking the value 1 if the MBG duration is equal to t, and δ_{SC} is a dummy representing a 30-day store credit return guarantee. The coefficients β_{+}, β_{-} represent the quadratic effects of return probability for MBGs longer and shorter than 30 days, respectively. Table 4 reports the results of the regression. Due to the nature of sample heterogeneity, we report standard errors using White's (1980) consistent correction.

The results in the table confirm Hypothesis 5, finding that a higher return probability increases the premiums for longer MBGs (positive signs for return probability, $t = 60, \infty$) and increases discounts for shorter MBGs (negative signs for return probability, $t = 0, 7$). Further, we find that the relationship is significantly nonlinear, with premiums concave in return probability for longer MBGs, and convex for shorter MBGs.

Table 4
Estimated parameters of discount/premium model.

Coefficient	In-store estimate ($)	Catalogue estimate ($)
Constant (t =0)	−9.570***	−11.054***
	(0.904)	(1.040)
30-day SC	5.460***	5.674***
	(1.034)	(1.266)
7-day MBG	4.705***	3.835***
	(1.091)	(1.289)
60-day MBG	10.813***	12.393***
	(0.989)	(1.141)
Infinity MBG	13.486***	14.938***
	(1.105)	(1.276)
RP ($t = 0$)	−0.162**	−0.233***
	(0.076)	(0.090)
RP ($t = 7$)	−0.192***	−0.355***
	(0.058)	(0.083)
RP, SC ($t = 30$)	−0.249***	−0.261***
	(0.070)	(0.089)
RP ($t = 60$)	0.121**	0.102**
	(0.055)	(0.057)
RP (t =∞)	0.127**	0.102
	(0.062)	(0.067)
RP^2 $t < 30$	0.003***	0.004***
	(0.001)	(0.001)
RP^2 $t \geq 30$	−0.002**	−0.001*
	(0.001)	(0.001)

Key: *P < 0.10, **P < 0.05, ***P < 0.01

Table 5
Predicted discounts and premiums compared with a 30-day MBG for average probability of return.

Coefficient	In stores ($)	With catalogs ($)
Discount, no MBG	10.90	14.00
Discount, 7-day MBG	6.50	12.44
Premium, 60-day MBG	2.26	2.89
Premium, infinite MBG	5.20	5.78
Discount, 30-day store credit	6.63	10.24

The negative coefficient on the interactive return probability and SC term indicates that customers with higher return probabilities also have higher valuations (higher premiums) for store credit refunds. This is consistent with the notion that customers with higher return probabilities should value return guarantees more.

The results from Table 4 can be combined with the average reported $\overline{RP}$ (10.13% in stores and 18.63% through catalogs) to obtain the predicted value for various-length MBGs. For example, the predicted discount for no MBG with average in store $\overline{RP}$ is $10.90, while for an individual with $\overline{RP}$ of 5% it would be $10.30. Remarkably, it appears that the effect of average $\overline{RP}$ on the valuation of the MBG is quite small (although significant). This again highlights the lack of ability of individuals to deal with probabilistic events, as they appear to only vaguely account for changes in the likelihood of using the MBG for a return. In calibrating the probability weighting function, Kahneman and Tversky (1979) note that probabilities smaller than 0.3 tend to be overstated in a way that would reduce the decline in perceived probability as the true probability tends towards zero. Thus, this result is very similar to that predicted by prospect theory. Table 5 contains predicted discount and premium values for the average return percentages.

From the model, we can calculate both the implied value of disappointment for a given discount/premium and stated return probability, as well as the change in return probability over MBG duration. Eq. (4) can be rearranged to find

$$D - R = \frac{\Delta P_t \pi(1 - \mathrm{RP}_t)}{\Delta\pi(\mathrm{RP}_t)} - P_{t_0}. \tag{8}$$

At $t = 0$, we know that $\Delta\pi(\mathrm{RP}_t) = \pi(\mathrm{RP}_{t_0})$, which is given by the stated RP. However, we have not calibrated the probability weighting function. Other than the probability weight, all variables in the right-hand side of (10) are known. If we consider return costs to be negligible, and if we know the probability weighting function, the calculation in (10) yields a reasonable estimate of the cost of disappointment aversion.

Furthermore, observing that

$$\pi(1 - \mathrm{RP}_t) = \pi(1 - \mathrm{RP}_{t_0} - \Delta\mathrm{RP}_t) = \pi(1 - \mathrm{RP}_{t_0}) - \Delta\pi(\mathrm{RP}_t) \tag{9}$$

we can rewrite Eq. (4) as

$$\Delta P_t = \frac{\Delta\pi(\mathrm{RP}_t)(P_{t_0} + D - R)}{\pi(1 - \mathrm{RP}_{t_0}) - \Delta\pi(\mathrm{RP}_t)},$$

which can be manipulated to yield

$$\Delta\pi(\mathrm{RP}_t) = \frac{\Delta P_t \pi(1 - \mathrm{RP}_{t_0})}{(P_{t_0} + D - R + \Delta P_t)}. \tag{10}$$

We can further manipulate (10) to obtain an estimate of the change in RP with respect to MBG duration:

$$\Delta\mathrm{RP}_t = \pi^{-1}\left(\frac{\pi(1 - \mathrm{RP}_{t_0})}{(P_t + D - R)}\Delta P_t + \pi(\mathrm{RP}_{t_0})\right) - \mathrm{RP}_{t_0} \tag{11}$$

Given our estimated $D - R$ from (8), we can use Eq. (11) to estimate how a change in MBG duration impacts the consumer's likelihood of returning the product.

Tables 6A and 6B contain calculations of both the value of disappointment and the change in consumer expected probability of return for various return periods predicted from our estimates given various starting values of return probabilities assuming the base case is a 30-day return MBG. These tables show that the cost of disappointment can be substantial for the range of return probabilities found in our data. Table 6A assumes that no weighting takes place (i.e., $\pi(\mathrm{RP}) = \mathrm{RP}$), and Table 6B assumes the probability weighting

Table 6A
Calibrated costs of disappointment and implicit expected customer's change in probability of return over time for various RPs assuming no probability weighting.

	In-store products				Catalogue products			
RP	0.05	0.10	0.15	0.20	0.05	0.10	0.15	0.20
Discount, no MBG ($)	10.31	10.89	11.33	11.61	12.12	12.98	13.65	14.11
Disappointment cost ($)	154.80	57.01	23.18	5.44	189.26	75.86	36.34	15.46
	Change in RP							
7-day MBG	−0.028	−0.060	−0.095	−0. 132	−0.037	−0.081	−0.130	−0.183
60-day MBG	0.009	0.021	0.035	0.050	0.008	0.018	0.029	0.043
Infinite-length MBG	0.022	0.048	0.078	0.114	0.018	0.038	0.060	0.085

Table 6B
Calibrated cost of disappointment and implicit expected customer's change in probability of return over time for various RPs using probability weighting.

	In-store products				Catalogue products			
RP	0.05	0.1	0.15	0.20	0.05	0.10	0.15	0.20
π(RP)	0.149	0.222	0.278	0.327	0.149	0.222	0.278	0.327
Discount, no MBG ($)	10.31	10.89	11.33	11.61	12.12	12.98	13.65	14.11
Disappointment cost ($)	35.21	6.28	-5.76	-12.94	48.62	15.38	1.47	-6.89
	Change in RP							
7-day MBG	−0.036	−0.077	−0.122	−0.170	−0.044	−0.094	−0.120	−0.200
60-day MBG	0.023	0.063	0.115	0.170	0.021	0.056	0.101	0.151
Infinite-length MBG	0.054	0.134	0.231	0.333	0.044	0.107	0.181	0.260
	Change in π(RP)							
7-day MBG	−0.150	−0.163	−0.171	−0.174	−0.256	−0.294	−0.326	−0.351
60-day MBG	0.039	0.045	0.049	0.051	0.039	0.045	0.049	0.052
Infinite-length MBG	0.092	0.097	0.101	0.105	0.089	0.092	0.095	0.098

function estimated in Tversky and Kahneman (1992), namely,

$$\pi^{+}(\mathrm{RP}) = \begin{cases} \frac{\mathrm{RP}^{\gamma}}{(\mathrm{RP}^{\gamma}+(1-\mathrm{RP})^{\gamma})^{1/\gamma}} & \text{if} \quad \text{“loss”} \\ \frac{\mathrm{RP}^{\delta}}{(\mathrm{RP}^{\delta}+(1-\mathrm{RP})^{\delta})^{1/\delta}} & \text{if} \quad \text{“gain”} \end{cases}$$

where $\gamma = 0.61$ and $\delta = 0.69$. In our case, a return is considered a loss, while a fit is considered a gain.

The results suggest that probability weighting (Table 6B) should accompany disappointment in any reasonable explanation of the survey responses. The results also indicate that for a 10% return rate, that which is most relevant for our sample, the net disappointment costs ($D - R$) from realizing a non-fit that cannot be returned is \$10.89. For the average range of return probability from the survey (about 10%), the predicted costs of a loss are much more reasonable when employing the weighting function at the same time, to get a value, for example, of \$6.28 instead of \$57 when the weighting return probability are not used. These finding supports Hypothesis 6. The multiplicative nature of probability in the value of the option means that a misperception in probability can greatly increase the option value of MBGs, even when the true return probability is not substantially increased. This makes MBGs an especially valuable tool to promote sales and possibly even profits.

The analysis also reveals that disappointment decreases with the probability of returning the product. This observation suggests that consumers who are averse to disappointments may check products more closely before deciding to purchase and would be therefore less likely to return them.

Comparison the model presented in our work against the model which has been the standard in the field—expected utility theory is one of the focal points of this paper. Throughout this paper we have used an expected net benefit model to approximate the expected utility model and demonstrated that it cannot explain the size of the observed valuation of an MBG in our sample. However, one might argue that it would be possible to find a concave utility function that might rationally explain the size of the discounts demanded for shorter MBGs. This is not the case. In order to demonstrate this, we can employ the calibration technique first introduced by Rabin (2000) and generalized by Just and Peterson (Just and Peterson, 2003, 2010). As Rabin noted a utility curve can be approximated locally by a straight line. In the context of expected utility this means that individuals should behave approximately risk neutral for small gambles. If individuals do not behave as risk neutral in small gambles, this behavior will often imply that the individual would display somewhat absurd levels of risk aversion for larger gambles if the utility function is concave. The basic problem is to find the utility function that minimizes the change in slope (minimizes the Arrow–Pratt measure of risk aversion) over the support of the gambles in the choice set, that rationalizes the selected gamble. In our case, the individual is indifferent between the 30-day MBG and no MBG. Just and Peterson (2010) show that the solution to this constrained optimization will always result in a utility function given by two connected line segments. Moreover, if the resulting line segments form an acute angle, then there is no monotonically increasing utility function that could reconcile the observed choice, and expected utility is clearly not applicable. Even if they do not form an acute angle, they may produce a very unlikely change in marginal utility. In Table 7, we report the angle of the two line segments. Anything below 90° is clearly infeasible (which is the case for each of the examples below). The angles remain relatively infeasible and are likely to result in negatively sloped portions of the utility function for larger return costs and widely different valuations of the product. Thus, expected utility seems to be an unsatisfactory explanation (see Appendix A2 for mathematical detailing).

Table 7
Minimum Change in Slope of the Utility function Implied by WTP for 30 Day MBGs.

	In-store products				Catalogue products			
RP	0.05	0.1	0.15	0.20	0.05	0.10	0.15	0.20
RC = \$0	66	68	70	72	58	60	62	64
RC = \$1	70	72	74	76	63	65	67	69

Anything below 90 implies that the function must contain a negative slope over some range of value.

5. Conclusions

This paper aims at explaining the anomalies observed in the price consumers are willing to pay for marketing risk reduction tools. We apply prospect theory by incorporating a loss aversion term to consumers' payoffs based on the possibility that the consumer does not receive the expected consumption benefits after purchasing the product because the product turns out not to fit the consumer's idiosyncratic needs. We demonstrate and test out theory on MBGs and show that "behavioral anomalies" such as aversion to loss and the distortion of probability beliefs play a role in the valuation of MBGs.

We use prospect theory to decompose the value of the MBG into three parts: (1) the "normal" returns accounted for in traditional analysis; (2) disappointment; and (3) overweighting the likelihood of return. The results of our analysis reveal significant deviation from expected net benefit valuation. The empirical results suggest that for this sample, disappointment may dominate other purchase risk considerations, inflating what otherwise might be a relatively modest cost for a non-fit.

This paper reveals the heterogeneous nature of demand for MBGs, where consumers vary in their valuations of the MBG, and relates this heterogeneity to differences in the likelihood of returning products. The combination of the high risk premiums that some consumers are willing to pay for an MBG option, together with heterogeneity in these valuations, suggests that retailers may profitably consider unbundling MBGs from products and offering the MBG as an optional item for customers to select at the time of purchase. This implication is consistent with that suggested in Heiman et al. (2002). Such a solution would permit customers to select the appropriate MBG depending on their prediction of whether they are more or less likely to return the purchased good. Customers could possibly even select an optimal return schedule depending on the product and their personal needs for continued assessment outside of the store. A similar strategy has already been successfully adopted in the context of warranties, with retailers and producers of electronic goods significantly increasing their profits by offering extended warranties and service contracts while reducing the basic warranties they offer (Stafford, 2006).

Store credit returns are less valuable than are cash refund MBGs. While this finding is not new, its importance to the retail sector is exemplified when taking into account that consumers consider disappointment costs and inflate low probabilities of a bad decision. The estimations suggest that consumers perceive that their expected loss in benefits when they are offered store credit instead of money is roughly similar to their loss in benefits from reducing the MBG duration from 30 days to 7 days, or over half of the value of the standard 30-day MBG. Thus retailers should be careful about substituting credit for cash in their return policies since this can seriously reduce the perceived value of the MBG.

We also find that MBGs have a greater value for catalog purchases than for in-store purchases, at least within the first 30 days. This finding is consistent with Ofek, Katona and Sarvary (2010), Bonifield, Cole and Schultz (2010), Zhang et al. (2010) and Shulman, Coughlan and Savaskan (2009). The large values attributed to disappointment from bad purchasing estimated in this paper provide a strong argument for why Internet retailing for experience goods has been much slower to gain ground than had been generally anticipated. Consumers' higher valuation of MBG for online shopping and lower value of store credit

may explain American Apparel returns policy that allow full refund for online purchase while offering only store credit for brick-and-mortar purchases.

These arguments raise several interesting research questions: Why do we observe a difference in product return policy between the US and the EU? Why is the product return policy in the EU[7] imposed by a law while in the US the standards are set by competition? Why is it that in the EU products cannot be returned if they were used? This restriction certainly restricts the ability of consumers to learn, thus limiting the value of the MBG.

Acknowledgements

We acknowledge the Asociación Mexicana de Cultura A.C. and the Davidson center for Agribusiness for their support.

Appendix A1

Let F indicates whether the product fits, where $F = 1$ indicates a product fit and $F = 0$ indicates a non-fit. Let M be the monetary cost associated with an outcome. In the context of having purchased a product with an MBG of duration t for a price of P_t, $M = 0$ if and only if the product is returned, and $M = -P_t$ otherwise. The third dimension is the return costs represented by effort E. If the consumer returns the product, this return effort reflects a loss of $E = -R$; otherwise $E = 0$.

From the previous notation, we can present the distribution of the three outcomes—fit, returned and disappointed—respectively, for the reference period as

Outcome: fit/nonfit	Monetary cost	Cost of return	Probability
1	$-P_{t_0}$	0	q
0	0	$-R$	$(1-q)f(t_0)$
0	$-P_{t_0}$	0	$(1-q)(1-f(t_0))$

where the outcomes are given on the left and their probability of occurrence are given on the right. Under a new duration period, the distribution of outcomes would be

Outcome: fit/nonfit	Monetary cost	Cost of return	Probability
1	$-(P_{t_0}+\Delta P_t)$	0	q
0	0	$-R$	$(1-q)[f(t_0)+\Delta f(t)]$
0	$M=-(P_{t_0}+\Delta P_t)$	0	$(1-q)[1-f(t_0)-\Delta f(t)]$

Since the utility of a change in the contract is defined on the difference between outcomes after adjustment, the distributions of the differences in the states of nature are

Changes in outcome: fit/nonfit	Changes in monetary cost	Changes in cost of return	Changes in probability of outcome
$\Delta F = 0$	$\Delta M = -\Delta P$	$\Delta E = 0$	q
$F = 0$	$M = 0$	$E = -R$	$(1-q)\Delta f(t)$
$F = 0$	$M = -(P_{t_0}+\Delta P_t)$	$E = 0$	$-(1-q)\Delta f(t)$
$\Delta F = 0$	$\Delta M = -\Delta P$	$\Delta E = 0$	$(1-q)[1-f(t_0)]$

where either the outcomes (on the left), or the probabilities of these outcomes (on the right), have changed. Since we are looking at changes in outcomes, the probabilities on the right no longer sum to one.

Following Kőszegi and Rabin (2007) who suggested that when the choice involves relatively small expenditure relative to income we define the valuation function (the equivalent of the utility function) in monetary terms. Under this assumption the monetary loss has the same weight as the gain, whereas the notion of loss aversion as presented in prospect theory suggests concavity in gains and convexity on losses. Concavity or convexity is captured in our model through $f(t)$ that yields non linear valuation functions. We assume that when $F = 0$ and $E = 0$, namely when the buyer is stuck with a product that is a non-fit, she incurs a disappointment cost of D, so that the gain from the reduced probability of being stuck with a non-fit product is included in the gain from avoiding disappointment cost together with the financial reimbursement when returning the product. We assume that return costs (R) are unaffected by the duration of the return period. With this modification, the distribution of the valuation of the change in outcomes is given by:

Changes in outcome	Changes in probabilities of fit/no-fit
$U[\Delta F = 0, \Delta M = -\Delta P_t, \Delta E = 0] = -\Delta P_t$	q
$U[F = 0, M = 0, E = -R] = -R$	$(1-q)\Delta f(t)$
$U[F = 0, M = -(P_{t_0}+\Delta P_t), E = 0] = -(P_{t_0}+\Delta P_t) - D$	$-(1-q)\Delta f(t)$
$U[\Delta F = 0, \Delta M = -\Delta P_t, \Delta E = 0] = -\Delta P_t$	$(1-q)[1-f(t_0)]$

Integrating the notion of the consumer's probability weighting function, π, into Eq. (6)[8] yields the consumer's willingness to pay for a marginal change in duration that is given by:

$$\Delta P_t = \frac{[P_{t_0}+D-R]\,\pi\left[(1-q)\,\Delta f(t)\right]}{\pi\left[q+(1-q)\left[1-f\left(t_0\right)-\Delta f(t)\right]\right]} = \frac{[P_{t_0}+D-R]\,\Delta\pi\,[RP_t]}{\pi\,[1-RP_t]} \tag{A1.1}$$

where, in the new notation given in the right term, RP represents the probability of returning the product and $\Delta\pi$[RP] represents the change in perceived probability of returning the product induced by the change in MBG duration. Eq. (7) suggests that the maximal (minimal) amount that consumers will be willing to pay (receive) for longer (shorter). MBG is a product of the increase (reduction) in the RP and the gain from returning an unfit product, divided by the probability of keeping the product.

Appendix A2

The 30-day MBG is a gamble with three outcomes: (i) the product fits, resulting in $V - P$. Let the utility of a fit under the zero-day MBG after paying for the good and receiving the compensating WTA_0 arbitrarily equal V, and the utility of being stuck with a non-fit (again after paying the price and receiving compensation) be equal to 0. Note that in this case, V must be above 30 if the individual is willing to purchase the good for \$40 with some probability of being stuck, or of having to return the item (the expected utility must be at least the utility equivalent of \$40). If we consider the move to a 0 day MBG the minimum concavity utility function will consist of two connected line segments, and must meet the constraint

$$\begin{aligned}(1-q)0 + q(V-P+\text{WTA}_0) &= -(1-f(30))(1-q)\beta\text{WTA}_0 \\ &+ f(30)(1-q)\beta(P-\text{WTA}_0+\text{RC}) + q[\beta(P-\text{WTA}_0+\text{RC}) \\ &+ \frac{V-\beta(-\text{RC}+P-\text{WTA}_0)}{V-P+\text{WTA}_0+\text{RC}}(V-P+\text{RC})\end{aligned}$$

[7] http://europa.eu/youreurope/citizens/shopping/shopping-abroad/returning-unwanted-goods/index_en.htm.

[8] $\Delta P_t = \frac{[P_{t_0}+D-R](1-q)\Delta f(t)}{[q+(1-q)[1-f(t_0)-\Delta f(t)]}$.

where β is the slope of the left part of the utility function, the slope of the upper part of the calibrated utility function is

$$\frac{V-\beta(-\mathrm{RC}+P-\mathrm{WTA}_0)}{V-P+\mathrm{WTA}_0+\mathrm{RC}}$$

The left hand side of the equation is the expected utility of the compensated MBG, and the right hand side is the utility resulting from the forgoing the purchase (0). This can be rewritten as

$$\beta=\frac{q\left[(V-P+\mathrm{WTA}_0)-\frac{V(V-P+\mathrm{RC})}{V-P+\mathrm{WTA}_0+\mathrm{RC}}\right]}{\left\{-(1-f(30))(1-q)\mathrm{WTA}_0+f(30)(1-q)(P-\mathrm{WTA}_0+\mathrm{RC})+q\left(P-\mathrm{WTA}_0+\mathrm{RC}-\frac{(-\mathrm{RC}+P-\mathrm{WTA}_0)}{V-P+\mathrm{WTA}_0+\mathrm{RC}}\right)\right\}}$$

Expected utility is apparently violated if the dot product of the vectors formed by each line segment is positive, or

$$[\beta(P-\mathrm{WTA}_0-\mathrm{RC})\ (P-\mathrm{WTA}_0-\mathrm{RC})]\times\begin{bmatrix}V-\beta(P-\mathrm{WTA}_0-\mathrm{RC})\\ V-P+\mathrm{WTA}_0+\mathrm{RC}\end{bmatrix}>0$$

or,

$$V(\beta+1)(P-\mathrm{WTA}_0-\mathrm{RC})-(\beta^2+1)(P-\mathrm{WTA}_0-\mathrm{RC})^2>0$$

In our data, $P=40$, WTA_0 is given by the survey, while the probability of return or being stuck with the product, $V$and RC are unobservable. For the purpose of simulation, we assume the probability of being stuck with the product in the 30 day MBG is 0.02 and use the observed average discount for both in store and catalogue purchases. We use a range of return costs, and assume $V=44$ (a 10% premium over the price).

Appendix B. Questionnaire

This questionnaire is part of a research study being conducted at UC Berkeley on the use of cash-back/money-back guarantees (MBG). We are interested in understanding how different people utilize MBG. Your participation will be very helpful to our study, and your answers will be kept confidential. We thank you for taking the time to participate in our survey.

1. On average, what percentage of clothes that you purchased in stores or catalogs do you return?
   Store: _____% Catalog _____% (Write N/A if not applicable.)
2. Are there certain items that you return more often? Please specify items_____
3. For an article of clothing (dress, trousers, etc.) purchased at a ***store*** and priced at $40 that comes with a 30-day money-back guarantee (MBG),
   How much off on this $40 item would you demand if:
   (a) it **did not come** with a MBG? $_____;
   (b) it came with only a 7-day MBG? $_____;
   (c) if you only received store credit or replacement instead of your money back? $_____
   How much more would you pay to upgrade the MBG to
   (a) 60 days? $_____
   (b) unlimited time? $_____
   For an article of clothing (dress, trousers, etc.) purchased from a ***catalog*** and priced at $40 that comes with a 30-day MBG.
4. How much off on this $40 catalog item would you demand if:
   (a) it **did not come** with a MBG? $_____;
   (b) it came with only a 7-day MBG? $_____;
   (c) if you only received store credit or replacement instead of your money back? $_____
   How much more would you pay to upgrade the MBG to
   (a) 60 days? $_____
   (b) unlimited time? $_____
5. What is your age? _____
6. What is your gender? Female____ Male____
7. What is your occupation? _____
8. What is your ethnicity? White____ Black____ Asian____ Hispanic____ Other____
9. Are you married? Yes____No____
10. Where do you live? _____
11. What is your average household income? Below $20,000_____; $21,000–$40,000_____; $41,000–$60,000_____; $61,000–$80,000_____; more than $81,000_____.

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